

Paraphernalia – Game Mechanics Facilitating Bodily Movement and Play

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ABSTRACT

This paper complements the article “Restrains as a Mechanic for Bodily Play” by presenting the paraphernalia of games as different mechanics that address the surrounding and contextual factors of movement-based game and play activities, while restraints address the players’ bodily preconditions. Based on an analysis of a collection of traditional games combined as bridging concepts, the mechanics are derived and exemplified in traditional and digital game exemplars and explained using theoretical concepts from phenomenology and postphenomenology. The presented mechanics provide a roadmap to design for and encourage bodily play by drawing on the historical development of (i.e., traditional) play and game activities and leveraging this knowledge into the domain of digital and technology-supported games and play activities.

CCS CONCEPTS

• Human-centered computing; • Interaction design; • Interaction design theory, concepts and paradigms;

KEYWORDS

Game mechanics, Movement-based games, Bodily play, Game Design, Play Design

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1 INTRODUCTION

This paper addresses two gaps in the literature on designing movement-based games by a) presenting a set of mechanics complementing Restrains as a Mechanic for Bodily Play [5] and b) leveraging knowledge from (non-digital) traditional movement-based games into the digital domain. While restraints [5] uncover designing for bodily play experiences through restrictions on bodily preconditions, the mechanics that facilitate and encourage the players to move are missing. Restrains [5] were derived from a

collection of traditional games addressing a gap in the literature by leveraging knowledge from (non-digital) game tradition into the digital domain. This paper complements the work on restraints by applying the same methodology, bridging concepts [2], to the same collection of games [7]. By following the same path, this paper investigates the mechanics that complement restraints by encouraging and facilitating movement.

With the same motivation as the authors of Restrains as a Mechanic for Bodily Play [5], this paper builds on the notion that before the era of digital games, (non-digital) games were, to a large extent based on the players’ physical movement. Because these game and play activities have been around - and refined - for millennia, we believe they contain helpful knowledge. Furthermore, few studies have sought to bridge these two domains of digital and non-digital game studies, and so this paper also addresses this same gap as the paper it complements. This investigation’s questions are, thus, which mechanics facilitate and support the players’ movement – and effectuate restraints? How can we leverage knowledge from traditional games into digital game design?

Concretely, four generic mechanics are presented and termed paraphernalia [16]. The word paraphernalia, borrowed from Cambridge Dictionary [16], is used here to describe the elements surrounding the player and influence the activity in different ways to effectuate restraints.

Practically, the collection of 140 traditional European games [7] was analyzed guided by theoretical concepts from phenomenology and postphenomenology following the bridging concept’s [2] methodology. The derived mechanics are further illuminated as conceptual contributions through exemplars using traditional games from the collection and subsequently demonstrated in two digital game examples. Because the collection of traditional games comprises more than 140 games, they are not listed here, but selected games (that were most suitable to demonstrate the argument) will be used as examples from the collection. The paper ends with a brief discussion and conclusion.

2 BACKGROUND

Restrains for bodily play in [5] is defined as; “*direct fixations or exclusions of body parts, or deprivation or manipulation of bodily senses instated by either rules, actions or the use of devices. When the players move, the restraints constitute unnecessary and arbitrarily chosen obstacles causing a change in the players’ bodily preconditions for actions and interactions, which leads to a difference in their bodily movement possibilities. For bodily play experiences, the change in the players’ preconditions for actions and interactions starts the process of reshuffling the equilibrium leading to kinetic joy rides*”. Scrutinizing this definition, restraints are defined as three types: *Fixations of body*

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parts, exclusions of body parts and deprivation or manipulation of bodily senses. Briefly summarized fixations of body parts are when, for instance, the players' heads are fixated to a balloon in balloon dance, or when the player shooting a penalty in handball has to keep their one foot fixated to the floor during the shot. Exclusion of body parts is when one or more of the players' body parts are not allowed to be used, e.g., hands in football (except the keeper) or feet in handball. We find deprivation or manipulation of bodily senses as a mechanic in, for example, Blind Man's Buff [15].

To uphold and practically infer restraints, the definition states that restraints are inferred rules, devices and actions. Restraints inferred by rules include, e.g., how the players in football are not allowed to use their hands for playing with the ball. Restraints inferred by devices are, e.g., how the PlayStation move controller restricts movement in *Johann Sebastian Joust* [3]. Lastly, restraints inferred by actions are, for instance, how players in physical fighting games hold on to each other and thereby infer restrictions on the other player's bodily preconditions for movement [5].

Theoretically, restraints are explained as leading to kinetic joy rides. Kinetic joy rides were introduced by Sheets-Johnstone [10, 11] as the synergy of movement sequences that form a whole – as an experience of play. To Sheets-Johnstone, movement is not experienced as singular motions but as sequences forming a whole. An example is walking as the sequence of repeatedly stepping forward alternately with our feet and legs. When movement sequences form a joy ride that reshuffles the player's equilibrium [6], we regard these to constitute a bodily play experience. The player's equilibrium is the player's bodily status quo, e.g., in balance. A way to encourage such movement sequences is to create unnecessary or arbitrarily chosen obstacles for the players to overcome – as Suits [13] and Caillois [1] have described play. Thus, the mechanics' role is to infer obstacles that reshuffle the player's equilibrium to open up for new and different movement possibilities and explore and develop movement sequences. Additional theoretical concepts include Suits' [13] explanation of resources allocated for play, Ihde's [4] multistabilities and Verbeek's [14] things as their "doing".

Suits [13] explains how - in play – players reallocate resources that were initially intended for instrumental purposes to the play activity. Similarly, in postphenomenology, are "things" argued to be understood as what they do – rather than their name or intended use [14]. A simple example of this argument is how a book will be a doorstopper when used to keep a door open – or an oddly shaped ball if used as such in, e.g., Dodgeball. In addition, Ihde [4] terms technologies (including both digital and non-digital technologies) multistabilities as they take on various roles depending on the situation at hand.

By understanding what the games' elements "do" and how they can take on different and other roles than the ones they were initially intended for, these concepts were used to derive the mechanics. Emphasis for the analysis was, thus, what – in terms of their doing – facilitates and supports the players' physical movement into kinetic joy rides. Concretely, the games' elements' "doing" were described and grouped into themes and affinities. This process derived the game mechanics below. They are described through examples and further illuminated through two digital exemplars.

3 PARAPHERNALIA

As explained in Cambridge Dictionary [16], the word paraphernalia is used here to describe the external elements that surround the player and influence the activity in different ways. As such, paraphernalia are concerned with all the elements needed for or connected with that particular activity [16]. From the analysis of the elements as their "doings", four different categories of paraphernalia were found:

- Collectables
- Demarcations
- Environmental Conditions
- Action Enablers

Briefly explained, *collectables* function as the activity's "call to action", i.e., they constitute a purpose for the player to move around. The players do so in spaces constituted by various *demarcations*. Furthermore, the players' bodily movements in the demarcated spaces are affected by *environmental conditions*. Sometimes, the players use an *enabler* to perform specific and particular bodily actions. While each category can consist of several types, not all types occur (or have to) in all games. Like restraints, each mechanic is implemented and upheld by rules, devices or actions and constituted by physical or conceptual elements.

3.1 Collectables

Collectables are game elements that, when "collected", lead to a progression or change in the activity. As such, collectables constitute a value, which can be both positive and negative. While collectables of positive value attract players, negatively valued elements are converted into items to avoid becoming a "non-collectable". An example is the game of catch, where the players avoid the catcher. The catcher becomes a "non-collect"¹. Accidentally "collecting" a non-collectable will yield a negative outcome like a disadvantage, damage or in other ways be of less value to the concerned players. The opposite, i.e., "collecting" a positive valued collectable, provides the player with a positive outcome like an advantage or valuable resource of a kind. Both positive and negative values lead to a progression in the activity, e.g., win/lose, score points, change team, etc.

"Collecting" refers to the way collectables are obtained or avoided in the game and requires bodily actions, e.g., the catcher *catches* one of the other players, the players *free* the other players, territories are *conquered*, and players *avoid* being caught. The actions to "collect" can be either protect, conquer, capture, hit, catch, touch, spot, choose, avoid, etc. For instance, in ball games, the players *protect* a "goal" from being *hit* while *conquering* the ball. Alternatively, they *protect* the ball while trying to *hit* the goal. In, e.g., hide and seek, the catcher *spots* another player who *hides*.

A game can also contain various "collectables", which can change value during the game. For instance, in ball games where both the ball and the goal are "collectables", the ball is a value that, when collected, yields a progression or change the state of the game: only when having the ball the team can score a goal. Another example

¹It can seem that any human ethical aspect is disregarded here, and humans are merely material things. However, this is not the view that is anticipated here. Instead, it is a view on game structures wherein the players are part of the structure itself. Therefore, ethics should always be considered a central part of playing games.

of how a ball changes value is Dodgeball, where the players avoid being hit by the ball. The ball then functions as a non-collectable. However, the players must also collect the ball and hit the other players. This way, the ball reverts between being a “collectable” and a “non-collectable”.

Lastly, collectables are constituted through different forms either as physical elements like cubes, balls, sticks, skittles, territory, etc. or as conceptually constituted elements like King or Princess, chickens, foxes, dragons, diamonds, coins, etc.

To sum up: A collectable is characterized by its *assigned value* coupled to an outcome or progression in the game, the *required actions* to “collect” it, and an identifiable *form* – (often stated by the rules):

- **Assigned value:** The value of the collectable in the game corresponds to how “collecting” the collectable enables or prevents achievement or progression in the game.
- **Required bodily action(s):** The specific bodily action determines how to “collect” or not the collectable.
- **Identifiable form or action:** The collectable’s physical form or concept enables the players to identify it among other elements.

3.2 Demarcations

Demarcations define the activity’s physical places and conceptual spaces by marking off fields, zones, goals, lines, etc. Examples of how demarcations mark off a physical place are found in the world of sports, e.g., football fields, tennis courts and track races. However, in some games collection, the physical places are not defined determinately but rather as conceptual spaces. As such, demarcations can also work to divide and define spaces. An example is how a simple line divides a space in two but does not define the size of the two spaces. For instance, in Border Fight [7], a line divides the space and two teams fight each other by dragging over the other team’s members to their side of the line. In this example, the line marks the border as a place while dividing the teams’ spaces. However, it does not mark the size of those. As such, demarcations create both physical places and conceptual spaces.

The spaces defined by demarcations are assigned different attributes linked to the activity like inside/outside, territories, start, end, goals, etc. These spaces can also function as collectables, e.g., goals or an end line determining a winner. Another example is *Kick the Can* [7], in which the can conceptually marks the place holding all the caught players – and where the not-caught players can free them.

Furthermore, the marked-off spaces are often attributed to different bodily actions. An example is the squares in a *Hopscotch* [7, 17] play grid. Besides defining the inside/outside of the grid, they also determine specific bodily actions: hop on one leg in single squares and two legs in double squares – while avoiding the square with the Hopscotch stone. Alternatively, in *Stealing Eggs* [7], the teams are safe in their own territory while they can get caught in the other teams’ territory. *Stealing Eggs* [7] is a game of catch where two teams fight to steal the other team’s eggs - without being caught. Each team has a home territory (in which they are safe). Their eggs are placed at the backline of that zone. Each team must steal eggs by entering the other team’s territory. At the same time, they have

to protect their own eggs. If a player is caught in the other team’s territory, they have to start anew from their own territory (and return the egg if they stole one). These examples demonstrate how game and play activities can include various spaces with different attributes.

To sum up, demarcations mark an activity’s physical place and conceptual space with different *attributes* and/or particular *bodily actions*. As such, demarcations and the space they create are recognized by their:

- **Assigned attributes:** Define the functions of the space like the teams’ territories, inside/outside, start or end of an activity, yield awards like points and goals, or divisions of space.
- **Allowed (or not-allowed) bodily actions:** The spaces are often defined by or defining for specific bodily actions as allowed – or not.
- **An identifiable form:** line, field, cans, zones, etc.

3.3 Environmental Conditions

Environmental conditions refer to conditions that are part of – or stem from - the particular environment surrounding the players in the specific place of the activity. These comprise weather conditions and physical facilities like grass, concrete, houses, trees, flooring, walls, furniture, light, sound or smell sources, etc. Typical for these conditions is that they create, augment or rectify the conditions for the activity.

The environmental conditions can be naturally occurring or artificially added. Naturally occurring conditions comprise gravity, trees, grass, water, snow, wind, daylight, moonlight, etc. While artificially added conditions comprise concrete, traffic, benches, furniture, etc. The characteristics of these elements are that they condition the player’s movement possibilities while they most of the time are unavoidable. Because they are unavoidable, these conditions are often fundamental to the games.

Following the above argument, some movement-based games or play activities require particular environmental conditions for the activity to be playable, like water for swimming, snow for skiing or ice for ice-skating. For instance, playing Hopscotch [17] requires a foundation on which the hopscotch stone can slide. Moreover, playing Hopscotch [17] at nighttime requires a light source – or bright moonlight. Additionally, playing *Stealing Eggs* [7] in a dung yard/mud puddle add other conditions than playing the same game on a grass field. While these conditions are naturally occurring, artificially added conditions can also have a fundamental role in games. For instance, an obstacle course primarily consists of artificially added and adjusted obstacles. Likewise, a chair can function as a shelter from the catcher in a game of catch. Alternatively, it can become a trap for the player that avoids being caught.

Also, music and rhythm form part of the environmental conditions. For instance, coordinating collective movements using singing and rhythm has been widespread in many traditional games. An example is *Blind Man’s Buff* [9], universally known as a game of catch where the catcher is blindfolded. In the Danish play tradition, this game has been a widespread party game based on a song, where the catcher is spinning around and only “freed” to catch the other players when the seventh line of the song is sung [9]. Also, in

Lap Chair [7], the players coordinate movement through singing. Lap chair is a social game where all players sit on each other's lap, forming a circle. When they succeed in forming the circle – and all players sit on the players' lap behind them, they start moving the circle round – and coordinating this activity, they sing while gradually changing movements. Moreover, singing has also been used by players to draw attention to themselves in, e.g., different games of catch and hide and seek [7].

To sum up, environmental conditions are elements found as either a natural part or artificially added to the environment and affect the players' movement possibilities. Environmental conditions are recognized as:

- **Elements that are part of the environment surrounding the activity and**
- **Condition the players' movement possibilities in different ways;** e.g., swimming, sliding, hiding, sailing,
- **Naturally occurring or artificially added elements,** e.g., weather conditions, housing, furniture, and bodily sense stimulations like light, sound and smell but also gravity belongs to this category.

3.4 Action Enablers

Action enablers refer to the devices necessary to perform specific bodily actions or movements, like bats to hit the ball in rounders, a ball to hit the other players in *Dodgeball* or sticks to hit the King in *Kubb* [7]. These (often mandatory) elements in the game activities make particular bodily actions possible. For example, pulling the rope in tug-of-war [7] enables the two teams to fight. Similarly, a racket in tennis enables the player to return the ball with more speed and power than without the racket. Another example is how woodblocks in *Go to Iceland for Herring* [7] enable the players to move towards *Iceland* without touching the ground. In *Go to Iceland for Herring* [7], two teams compete to reach *Iceland*, and while doing so, they can place a woodblock for each turn. *Iceland* is a marked off place located at a distance from the start line – resembling an island in the sea. Players are not allowed to touch the ground in the sea, i.e., from the start line to *Iceland*. Instead, they can use their woodblocks as enablers to reach farther. Furthermore, when there are several woodblocks (they can use each other's woodblocks as pillars), they can form bridges of team members by crawling on top of each other. Also, activities like balancing on a slackline use action enablers like, e.g., a stick to support balance.

The above examples demonstrate how action enablers provide the players with extra resources to achieve actions and create movement possibilities that they are not otherwise capable of doing. Action enablers are characterized by elements that *enable specific bodily actions*. As was demonstrated, the outcomes enabled through “action enablers” comprise reaching goal spaces from afar with, e.g., a ball, or a bridge of people on woodblocks, gathering the strengths of many as in tug-of-war [7], or keep balance using a stick. This way, various elements can work as action enablers and provide the player with the means to perform actions and reach outcomes they would not otherwise be capable of.

While enabling some actions, enablers can also prevent or limit other actions. For example, a racket enables returning a ball quickly with high speed. However, it is not as useful for, e.g., climbing.

Nevertheless, climbing with a racket might be fun in other ways as it will somewhat restrict movement, i.e., impose a restraint. As such, an enabler might, in one instance, aid the player to do a specific action while in others might impose a restraint.

To sum up; Action enablers are digital or analogue technologies that enable the players to perform movements that they would not otherwise be able to do. Action enablers are recognized as technologies that:

- **Enable specific bodily actions,** e.g., rackets, sticks, controllers, etc.
- **It can also prevent specific bodily actions while enabling others.**

4 PARAPHERNALIA IN (DIGITAL) ACTION:

In the following section, two digital games are analyzed through the above-described mechanics to illuminate how they encourage the players to move in different ways. These examples also highlight how traditional games bear a resemblance to and thereby can inspire the design of digital games.

4.1 Johann Sebastian Joust

Johann Sebastian Joust [3] is a multiplayer game based on Johann Sebastian Bach's Brandenburg Concertos and is played using a PlayStation console and the accompanying Move Controllers. The controller's sensitivity is linked to the rhythm of the music, making them very sensitive to the tempo of the music. When it is slow, the threshold for sensitivity is low, meaning that the controller is triggered by little disturbance and the player is out of the game. When the tempo increases, the threshold rises and the controller can resist more disturbance without triggering. The game objective is that the last player with a not-triggered controller wins. The players have to move around protecting their controller while disturbing the other players in ways that make them move their controller above these thresholds and “die”.

This game is based on music – as an environmental condition. The move controller works as different elements: it restricts the players' preconditions for movement, hence working as a restraint while becoming collectable to “collect” or protect. In the endeavors to do so, the players move around a designated place determined by the range of the technology (as demarcations) while adhering to other environmental conditions such as furniture, stairs and other players.

The players can take on various functions interchangeably as they are collectables and non-collectables simultaneously. Moreover, they can also emerge as environmental conditions, e.g., as a kind of shelter from other players when they move around trying to avoid each other.

Together, these various elements (restraints and paraphernalia) form unnecessary and arbitrarily chosen obstacles. The obstacles encourage the player to explore different movement sequences, which reshuffles the players' equilibrium. Consequentially, overcoming these obstacles can lead to kinetic joy rides as the movement sequences that each obstacle encourages can merge in synergy. For instance, while protecting the move controller, the player also has to avoid another player and a chair simultaneously.

4.2 Space Agent

Space Agent [12] is a sound-based pervasive game where the players capture invisible aliens using their smartphones. The players search the physical space for the aliens in a similar way to *Pokémon Go* [8]. However, the aliens are invisible and only audible in a 3D binaural soundscape. To capture the aliens that attack from a random location in a 360-degree sphere around the player, the player makes specific gestures with the smartphone while pointing in the alien's direction.

There is no visual representation of the aliens in this game – only audible. Because the main objective is to capture the invisible aliens, the main mechanic is the deprivation of the visual sense as a restraint. The aliens are the collectables that the players search to capture, and the players are non-collectables as they avoid being hit by the aliens. The smartphone works as an enabler as it enables the players to do the movements that capture the aliens in the digital world.

There are no demarcations in *Space Agent* [12] as the game is pervasive. However, plenty of environmental conditions affect the players' possibilities for moving, like traffic, houses, benches, signs, not to mention weather conditions, e.g., rain, snow, temperature, etc. Moreover, while enabling the capturing movements, the smartphone also restrains the player's hand to the smartphone – like the players' hands were fixated to their move controllers in *Johann Sebastian Joust* [3].

These examples point to how the different elements can infer various mechanics – at the same time. While the smartphone works as an enabler, it also imposes a restraint on the player as it fixates the one hand. Additionally, the aliens are both collectables and obstacles to avoid. Furthermore, the players have to navigate the virtual space auditorily while simultaneously navigating the physical space visually. These contradictions form obstacles to overcome as their bodily senses are deprived alternately for each domain. From a postphenomenological perspective, these elements form multistabilities. The following discussion section further elaborates on these perspectives.

5 DISCUSSION

Answering the questions posed in the introduction, this paper has presented the following four mechanics facilitating and supporting movement: Collectables work as the game's "call to action" as it entails movement in itself. Players cannot collect, capture, protect, conquer, shoot, find or reach anything without moving. As such, collectables *facilitate* movement while the other three mechanics *support* movement in various ways. Demarcations define the spaces for play, environmental conditions determine the premises for movement, and action enablers enable specific movements. These mechanics complement restraints in forming the conditions for kinetic joy rides and, consequently, bodily play and game experiences. Together restraints and paraphernalia address the primary conditions for bodily play as bodily preconditions and surrounding conditions. However, the way they do so is up to the designers to arrange – and the players to perform.

As stated in the introduction, restraints do not make a game, nor do any mechanics presented here separately. They work in synergy to form a game's core activity. Because a restraint is only effectuated

when the players move, the mechanics in paraphernalia are needed to encourage and define the conditions for movement. However, not all paraphernalia or restraints are represented in every movement-based game. While each mechanic can be designed and implemented individually – like a restraint – rarely can such elements encourage play in themselves. Like the restraint of not moving the move controller in *Johann Sebastian Joust* [3] has no effect if it is not part of the more extensive system of mechanics manifested in different elements, i.e., music, other players and physical space and things therein. The argument here is concerned with the synergies between the mechanics. While such concern is a critical focus of any game design, the emphasis is on the players' physical movement. If, for instance, we removed the rule that the last person with an active move controller wins in *Johann Sebastian Joust* [3], the players would have nothing to collect or protect. What then encourages the players to move around? Or, if the demarcations in *Hopscotch* [7, 17] were altered or removed, how would that affect the players' physical movement?

The analysis made it apparent how the elements changed role and "doing" and constituted different mechanics. For such discussion, we want to recall Suits' [13] explanation of how resources in play are reallocated to suit the situation and the postphenomenological perspective on technologies as multistabilities [4] and as doings [14]. In the examples, we saw how the smartphone functioned as an action enabler while inferring a restriction on the players' bodily preconditions. Also, the players in *Johann Sebastian Joust* [3] became collectables, non-collectables and environmental conditions simultaneously – and so did the aliens in *Space Agent* [12]. This way, we can say that the elements changed doings interchangeably throughout a game. Thus, the doing of a thing as a technological or human resource for play emerges as a consequence of both the thing as a doing and the player's need in the situation. A resource can perform as many doings simultaneously – as multi-doings constituted in relation to the player.

While the mechanics presented here encourage different – and possibly playful – movements, whether they lead to kinetic joy rides is up to the player. Furthermore, because a game experience is the player's subjective experience, so are kinetic joy rides. Therefore, the mechanics presented here can only help designers design for such experiences while the experiences remain the players'.

Finally, the above list does not represent a hierarchy between the mechanics. Instead, we advocate seeing the different mechanics as generative resources that can be added and removed as desired by the players – as they are made possible and arranged by the designers.

6 CONCLUSION

This paper has explained a game's paraphernalia as game elements and their functions as mechanics to constitute a game's core activity explained as unnecessary and arbitrarily chosen obstacles to overcome. The empirical basis was a collection of 121 *Euro Games* as traditional movement-based games. This paper and investigation complement the paper "*Restraints as a Mechanic for Bodily Play*". While the complemented paper formulates an essential contribution to the community, we have argued that it needs to be considered together with a game's paraphernalia, i.e., elements surrounding

the player and constituting a game's obstacles to overcome. By applying the same methodological procedure as the complemented paper, this paper has presented four game mechanics to encourage movement and complement restraints: Collectables – a game's "call to action", demarcations to define the space, environmental conditions that influence the players' movement possibilities, and action enablers that enable the players to do actions that would otherwise not be possible. These findings were described using game exemplars from the collection and further contextualized in two diverse digital game examples. The latter was done to leverage the derived knowledge from traditional games into interactive and digital technology-supported games. However, the presented contributions and examples are by no means exhaustive. Nevertheless, we believe they provide an insight into traditional movement-based games.

With this work, we hope to inspire researchers and designers of movement-based games and play activities to advance this field of study further.

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